

Exploring Logit Response Equilibria

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Outline

- ▶ Bounded Rationality in the Travelers' Dilemma
- ▶ Generalized Matching Pennies
- ▶ Alternative structural approaches
- ▶ Linear Probability Model
- ▶ regular QRE

The Travelers' Dilemma

- ▶ The dilemma due to Basu (AER, 1994)
 - ▶ Players: $i \in \{1, 2\} = I$
 - ▶ Strategies: $s_i \in \{80, 81, 82, \dots, 199, 200\} = S_i$
 - ▶ Payoffs: $\pi_i(s_i, s_j) = \begin{cases} \min(s) + r & \text{if } s_i < s_j \\ \min(s) & \text{if } s_i = s_j \\ \min(s) - r & \text{if } s_i > s_j \end{cases}$
- ▶ Unique Nash Equilibrium is $\{80, 80\}$ for all $r > 1$
- ▶ Intuitively people's behavior will be sensitive to r
 - ▶ when r is large people's behavior will be close to the Nash Equilibrium
 - ▶ when r is close to 1 people's behavior will be close to 200, that is, as far away from NE as possible
- ▶ The actual outcome is an empirical question.

Capra, Goeree, Gomez, and Holt (AER, 1999)

► Design

- 10 to 12 subjects randomly paired at the start of each round
- 10 rounds with the same r , penalty parameter
- Claims were required to be between \$0.80 and \$2.00
- Six sessions were run, each with a different penalty parameter

► Results

- $r = \$0.80$
 - round 1: claims average \$1.20
 - rounds 7-10: claims average \$0.80
- $r = \$0.05$
 - round 1: claims average \$1.80
 - rounds 6-10: average claims are close to \$2.00

Logit Response Equilibrium

- ▶ Ignore the apparent learning for now.
- ▶ Logit response equilibrium is sensitive to r , which indexes the optimization premium.
- ▶ See Excel worksheet.

Generalized Matching Pennies

- ▶ Ochs (GEB, 1994) tested the mixed strategy equilibrium prediction that players strategies depends on their opponents payoff and not their own by studying Generalized Matching Pennies.
 - ▶ Found behavior was sensitive to ones own payoff.
 - ▶ McKelvey and Palfrey (GEB, 1995) show that Logit Response Equilibrium can explain this sensitivity.
- ▶ Goeree, Holt, and Palfrey (GEB, 2003) reexamine the question.

Two Generalized Matching Pennies Games

- Consider the following pair of 2×2 generalized matching pennies games:

$$g(360) = \begin{array}{cc} & \begin{array}{cc} \textit{Left} & \textit{Right} \end{array} \\ \begin{array}{c} \textit{Top} \\ \textit{Bottom} \end{array} & \begin{array}{cc} 360, 36 & 36, 72 \\ 36, 72 & 72, 36 \end{array} \end{array} \quad \begin{array}{c} p \\ (1-p) \end{array}$$

$q \qquad (1-q)$

$$g(40) = \begin{array}{cc} & \begin{array}{cc} \textit{Left} & \textit{Right} \end{array} \\ \begin{array}{c} \textit{Top} \\ \textit{Bottom} \end{array} & \begin{array}{cc} 40, 36 & 36, 72 \\ 36, 72 & 72, 36 \end{array} \end{array} \quad \begin{array}{c} p \\ (1-p) \end{array}$$

$q \qquad (1-q)$

- Row player should not respond to the change in her payoffs as switch from $g(360)$ to $g(40)$, because column player payoffs have not changed.
- It is the column player who should respond even though his payoffs have not changed.

Nash and Logit Response Equilibria for $g(360)$: $\lambda = 0.1$

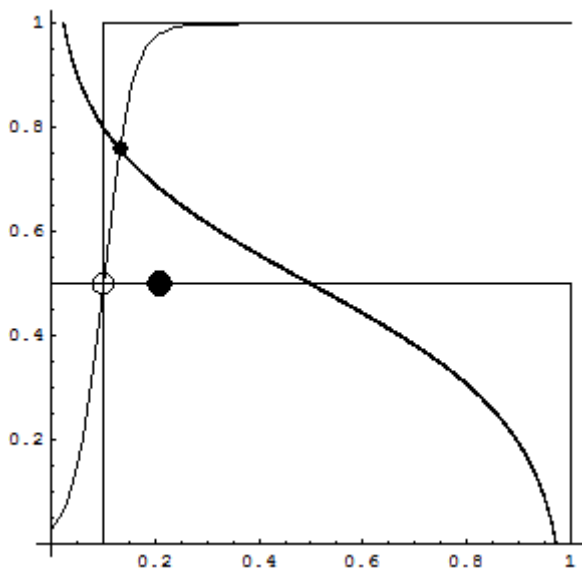
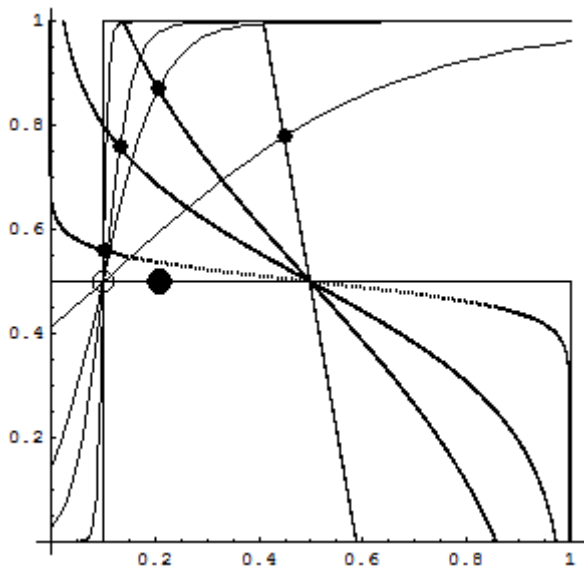


Figure : Nash and Logit Response Equilibrium for $g(360)$.

Nash and Logit Equilibria for $g(360)$:
 $\lambda = \{0.01, 0.05, 0.1, 0.5\}$



Nash and Logit Equilibrium for $g(40)$: $\lambda = 0.1$

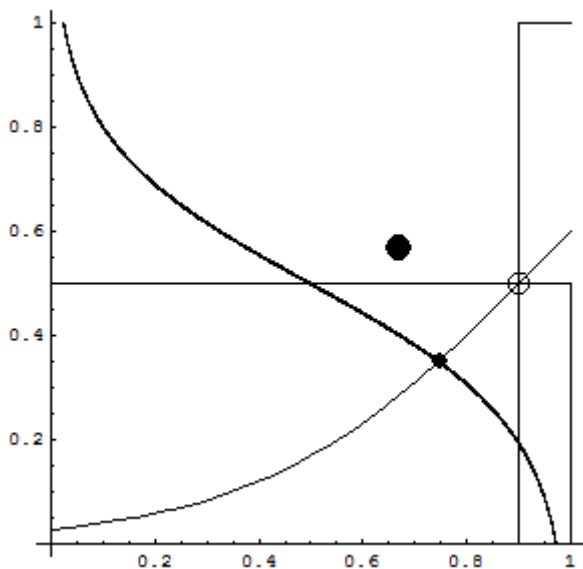
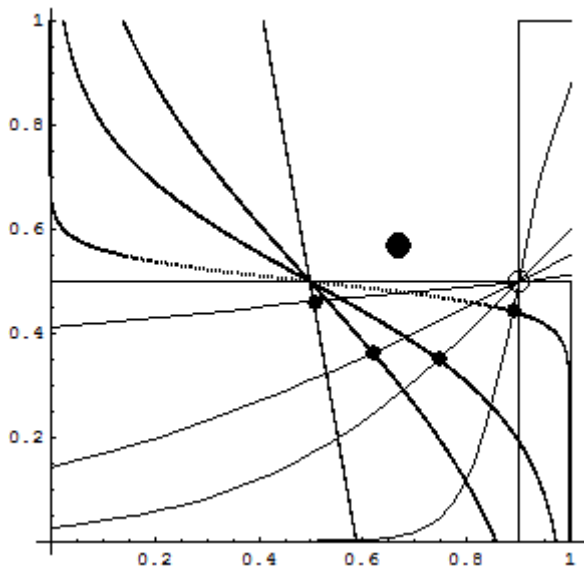


Figure : Nash and Logit Equilibrium for $g(40)$

Nash and Logit Equilibria for $g(40)$:
 $\lambda = \{0.01, 0.05, 0.1, 0.5\}$



Alternative Structural Approaches to Binary Choices

- ▶ Sometimes the one parameter Logit Equilibrium can not be fit to the data, see previous slide.
- ▶ Therefore consider alternative structural approaches.
- ▶ Recall that utility is stochastic in the structural approach

$$U_i \equiv u_i + \varepsilon_i,$$

- ▶ Where ε_i is a stochastic variable for the attractiveness of choice i ;
 - ▶ And u_i is the normal utility function for choice i .
- ▶ The probability that alternative i is chosen is

$$P_A(i) = \Pr(U_i = \max_{j=1,2} U_j)$$

- ▶ Let $f(x)$ be the density of ε ,
- ▶ where $\varepsilon \equiv \varepsilon_2 - \varepsilon_1$
- ▶ then $P_A(1) = \int_{-\infty}^{u_1 - u_2} f(x) dx$

Linear Probability Model

- ▶ Let ε be uniformly distributed on the interval $[-L, L]$ then

$$f(x) = \begin{cases} \frac{1}{2L} & \text{if } x \in [-L, L] \\ 0 & \text{otherwise} \end{cases}$$

- ▶ The choice probability for alternative 1 is

$$P_A(1) = \begin{cases} 0 & \text{if } u_1 - u_2 < -L, \\ \frac{u_1 - u_2}{2L} + \frac{1}{2} & \text{if } -L \leq u_1 - u_2 \leq L, \\ 1 & \text{if } L < u_1 - u_2 \end{cases}$$

Haile, Hortacsu, Kosenok (AER, 2008)

- ▶ Even with strong a priori restrictions on unobservables, QRE imposes no falsifiable restrictions
- ▶ that is, it can rationalize any distribution of behavior in any normal form game.
- ▶ Usually, in applications ε_{ij} are assumed to be independently and identically distributed across all choices j
- ▶ Furthermore, in most applications every ε_{ij} is an independent draw from an extreme value distribution giving the logit model.
- ▶ Relax the i.i.d. restriction and you can fit any observed behavior.
- ▶ Any falsifiable implication of QRE derives from additional maintained hypotheses on payoff perturbations, like i.i.d.
- ▶ The i.i.d. assumption is too restrictive
 - ▶ to fit a rich dataset
 - ▶ to fit observed comparative statics
 - ▶ to lead to reasonable out-of-sample predictions

Regular QRE by Goeree, Holt, and Palfrey (EE, 2005)

- ▶ Let P_{ij} denote the probability that player i selects strategy j

Definition

$p_i : R^{J(i)} \rightarrow \Delta^{J(i)}$ is a *regular quantal-response function* if it satisfies the following four axioms:

1. Interiority: $P_{ij}(\pi_i) > 0 \forall j = 1, \dots, J(i)$ and $\forall \pi_i \in R^{J(i)}$
2. Continuity: $P_{ij}(\pi_i)$ is a continuously differentiable function $\forall \pi_i \in R^{J(i)}$
3. Responsiveness: $\frac{\partial P_{ij}(\pi_i)}{\partial \pi_{ij}} > 0 \forall j = 1, \dots, J(i)$ and $\forall \pi_i \in R^{J(i)}$
4. Monotonicity: $\pi_{ij} > \pi_{ik}$ implies that $P_{ij}(\pi_i) > P_{ik}(\pi_i) \forall j, k = 1, \dots, J(i)$

Regular Quantal Response Equilibrium

Definition

Let P be regular. A *regular quantal response equilibrium* (R-QRE) of the strategic form game G is a mixed strategy combination σ^* such that

$$\sigma^* = P(\sigma^*).$$

Theorem

There exists a Regular Quantal Response Equilibrium of G for any regular P .

Proof.

Follows from Brouwer's fixed-point theorem.



Back to Luce

- ▶ Sometimes a power law fits data with large scale changes better than the Logit Equilibrium.

$$P_{ij} = \frac{(\pi_{ij})^{\frac{1}{\mu_i}}}{\sum_{k=1}^{J(i)} (\pi_{ik})^{\frac{1}{\mu_i}}}$$

- ▶ Laury and Holt (2002)
- ▶ Goeree et al. (2002)